

Education

University of Toronto

September 2024 - December 2025 (Expected)

MSc in Applied Computing (MScAC), Applied Mathematics stream.

2024 Vector Scholar in Artificial Intelligence (\$17,500)

Coursework: Generative NN, Nonlinear Numerical Analysis, Physics-Based Animation, Physics-Informed Neural Representations for Visual Computing, Communication for Computer Scientists, Technical Entrepreneurship.

University of Toronto

September 2020 - June 2024

HBSc Mathematics and Physics Specialist. High distinction.

Coursework: Machine learning, Deep learning, numerical analysis, optimization, real analysis, complex analysis, PDEs, linear algebra, Fourier analysis, probability, topology, differential geometry, nonlinear physics, quantum information theory.

Experience

1. Deep Learning Researcher · Vector Institute

November 2023 - Present

- First author publication in ICLR 2025. [Open Review](#).
- Conduct research in interpretability through extensive empirical experimentation and theoretical exploration on the transformer architecture.
- Discovered novel phenomenon coined Transformer Block Coupling, displayed positive correlation with generalization.
- Performed extensive experimentation on hundreds of open source transformer-based models.

2. Algorithms Researcher · University of Toronto, Department of Mathematics

April 2023 - Present

- Developing algorithms for solving scalar differential equations and for numerical integration.
- Our methods dominate over alternative methods in the high-frequency regime.
- Independently decreased time complexity by a factor of n by exploiting an over-determined system.
- Proved algorithm correctness by establishing theoretical bounds on computation error.
- Establish theoretical framework, implement, test and iteratively optimize, publish results in academic journals.

3. Teaching Assistant · University of Toronto

September 2023 - Present

- Run tutorials, host office hours, mark and invigilating exams.
- Winter 2025: [APM348](#), Mathematical Modeling.
- Winter 2025: [CSC336](#), Numerical Methods.
- Fall 2024: [MAT292](#), ODEs for second-year engineering students.
- Winter 2024: [MAT187](#), Calculus 2 for first-year engineering students.
- Fall 2023: [MAT244](#), ODEs for second-year math major students.

4. Machine Learning Researcher · DUNE × University of Toronto

September 2023 - April 2024

- Trained a graph neural network to detect tau neutrinos and their decay products.
- Combatted an imbalanced dataset by implementing a modified, weighted loss function.
- Improved classification performance on underrepresented training samples by over 20%.

5. Tutor · Forest Hill Tutoring

April 2023 - August 2024

- High school math and physics tutoring, largely focused on helping grade 11 and 12 IB students.
- Principal instructor for Ontario credited courses such as MDM4U, MHF4U and MCV4U.

6. Structural Engineering Assistant · Safeline Construction Group

April 2022 - October 2022

- Worked on a 57 storey building in downtown Toronto assisting the senior and assistant supervisors.
- Independently managed interior renovation of a small townhouse in downtown Toronto.
- Worked alongside mechanical, civil, and electrical engineers throughout entire construction process.

Awards, Publications & Talks

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| • Vector Scholarship in Artificial Intelligence. | 2024 |
| Awarded by the Vector Institute . Masters entrance scholarship valued at \$17,500. | |
| • Transformer Block Coupling and its Correlation with Generalization in LLMs. <i>ICLR</i> | 2025 |
| Murdock Aubry, Hoaming Meng, Anton Sugolov, Vardan Papyan. Work affiliated with experience (1). PDF . | |
| • A Solver for Linear Scalar ODEs Whose Running Time is Bounded Independent of Frequency. <i>Preprint.</i> | 2023 |
| Murdock Aubry, James Bremer. Work affiliated with experience (2). PDF . | |
| • On the Two Dimensional Levin Method. <i>Preprint.</i> | 2024 |
| Murdock Aubry, Kirill Serkh, James Bremer. Work affiliated with experience (2). PDF . | |
| • The Emergence of Clusters in Self-attention Dynamics. <i>Talk.</i> | 2023 |
| UofT Graduate Applied ² Math Seminar. Work affiliated with experience (1). Slides . | |
| • Development of Graph Neural Networks for Tau (τ) Neutrino Detection. <i>Talk.</i> | 2023 |
| CERN and Deep Underground Neutrino Experiment (DUNE). Work affiliated with experience (4). Slides . | |
| • On the Adaptive Levin Method. <i>Talk.</i> | 2023 |
| Canadian Undergraduate Mathematics Conference. Work affiliated with experience (2). Slides . | |

Technical Skills

- **Languages.** Python, MATLAB, Fortran, HTML, CSS, SQL, Bootstrap, Bash, Mathematica, L^AT_EX, Markdown.
- **Packages.** PyTorch, Transformers, TensorFlow, Numpy, Scipy, CUDA, Pandas, Matplotlib, Seaborn, JSON, OS, Selenium, BeautifulSoup, ezdxf, dxfgrabber.
- **Software.** Git, SSH, SLURM, Tensorboard, AutoCAD, Unix.